

Master Patient Index (MPI): A Technical and Operational Treatise

Target Audience: Health IT Analysts, Architects, and Developers

1. Introduction: The Identity Hypothesis

In healthcare, patient identity is not a static fact; it is a resolvable, auditable hypothesis. Clinical systems (EMRs, LIS, RIS) are designed to manage **records**—sets of clinical observations, orders, and billing events—using a local identifier (MRN). However, these systems are notoriously poor at managing **identity**—the mapping of multiple disparate records to a single human being.

The fundamental challenge of a Master Patient Index (MPI) is not just "finding duplicates," but managing the lifecycle of identity resolution without compromising clinical safety, legal integrity, or global health equity.

The Golden Rule of MPI

Patient IDs represent records. Person IDs represent reality.

2. The Reference Architecture: Three-Layer Separation

Successful MPIs rely on a fundamental separation of concerns. Collapsing identity truth into a search index is the most common cause of MPI failure.

Layer 1: The Identity Anchor (person)

The person table is the stable, long-lived anchor. It contains no clinical data and rarely changes. It serves as the enterprise-wide "Source of Truth" for human identity.

Layer 3: The Identity Assertion (person_patient_xref)

This layer maps person_id to various patient_ids across different systems. It records the **judgment** that "Record A and Record B belong to Person X." It must be auditable, reversible, and account for **Source System Trust**—weighting identifiers from national registries higher than self-registration portals.

Layer 3: The Search Engine (mpi_patient_index)

The **perception** layer. It contains denormalized, cleaned, and phonetic versions of demographics. It is optimized for speed and recall, identifying *candidates* for review.

3. Technical Implementation: The Schema

Database Design (MySQL 8+ Example)

The following schema enforces transactional safety and clear ownership boundaries.

```
-- Identity Anchor
CREATE TABLE person (
  person_id CHAR(36) PRIMARY KEY, -- Use UUIDs for future HIE federation
  status ENUM('active','retired','merged','error') NOT NULL DEFAULT 'active',
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);

-- Cross-Reference (The Assertion Layer)
CREATE TABLE person_patient_xref (
  person_id CHAR(36) NOT NULL,
  patient_id VARCHAR(32) NOT NULL,
  source_system VARCHAR(20) NOT NULL,
  source_trust INT DEFAULT 50, -- 0-100 weighting
  link_reason ENUM('registration_new','registration_selected','mpi_auto','mpi_manual','system_import') NOT NULL,
  status ENUM('active','superseded','revoked') NOT NULL DEFAULT 'active',
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (person_id, patient_id),
  FOREIGN KEY (person_id) REFERENCES person(person_id)
);

-- Search Index (The Perception Layer)
CREATE TABLE mpi_patient_index (
  patient_id VARCHAR(32) PRIMARY KEY,
  name_1_canon VARCHAR(100), -- Global-friendly (e.g. Given)
  name_2_canon VARCHAR(100), -- Global-friendly (e.g. Family)
  full_name_norm VARCHAR(300),
  alt_name_keys JSON,
  phonetic_code VARCHAR(50),
  dob DATE,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

4. Global Name Handling: The Five Representations

In global contexts, "First/Middle/Last" structures often fail. A robust MPI uses generic name components (Name 1, Name 2, etc.) to handle patronymics, double surnames, or mononymous patients.

1. **RAW:** Exactly what was typed (Dr. John O'Malley).
 2. **CANONICAL:** Cleaned for human review (JOHN OMALLEY).
 3. **NORMALIZED:** Collapsed strings (JOHNOMALLEY) for fast B-tree indexing.
 4. **ALTERNATE ORDER:** Defense against field swaps (OMALLEYJOHN).
 5. **PHONETIC:** Algorithm-generated codes (JNMAL). Must use language-appropriate algorithms (e.g., Metaphone for English, custom for Arabic/Chinese).
-

5. Matching Theory: Fellegi-Sunter & Source Trust

MPI matching uses **Fellegi-Sunter (F-S) Theory** to produce a **Total Score** based on weighted evidence.

- **M-probability:** Likelihood fields agree given a match (e.g., National ID).
 - **U-probability:** Likelihood fields agree by chance (e.g., common surnames).
 - **Source Trust Weighting:** Identity decisions are weighted by the source system's reliability. A link asserted by a National Identity Registry carries more weight than one from a legacy clinic spreadsheet.
-

6. Interoperability & Standards (HL7/FHIR)

Internal identity decisions must propagate using international standards to maintain semantic interoperability.

- **FHIR:** Map internal links to `Patient.link`. Use `replaced-by` for merges and `seealso` for high-confidence candidates.
 - **HL7 v2:** Use the MRG (Merge Patient) segment in ADT messages to inform downstream systems (LIS, PACS) of identity resolution.
-

7. Practical Implementation: The "In-App" Enterprise Pattern

When a separate MPI service is not feasible, the MPI can be built directly within the enterprise database using a dedicated schema to maintain logical separation.

The Architectural Strategy

```
CREATE SCHEMA identity_mpi;
```

```
CREATE TABLE identity_mpi.persons (  
  person_id      UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  canonical_mrn  VARCHAR(64) UNIQUE,  
  global_status  ENUM('active', 'merged', 'suppressed') DEFAULT 'active'  
);
```

```
-- Link existing EMR records to the Anchor
```

```
ALTER TABLE public.patients  
ADD COLUMN enterprise_person_id UUID REFERENCES identity_mpi.persons(person_id);
```

Key Advantages

1. **Transactional Integrity:** Wrap merge/unmerge in a single ACID transaction.
2. **Zero Latency:** Perform direct JOINS for real-time duplicate detection.
3. **Atomic Backups:** Clinical and identity truth stay perfectly synchronized in snapshots.

8. Operational Stewardship

Even the best algorithm requires a human **Data Steward**.

- **Stewardship Workflow:** Provide a UI for adjudicating "Medium Score" candidates.
- **The "Blast Radius" Principle:** Merging and unmerging should only affect the xref layer. **Never delete clinical history or merge records in clinical tables.**

9. Conclusion

A scalable MPI is a combination of a high-performance search engine and a rigorous identity layer. By treating identity as a resolvable hypothesis and adhering to global standards, Health IT organizations can ensure patient safety and data continuity across diverse ecosystems.

Indexes find candidates. Links assert truth. Persons preserve reality.